MILESTONE 6 · FINAL PROJECT PRESENTATION

Course-Aware Personalized Learning Companion - Machine Learning

Retrieval-Augmented Q&A · Question Intelligence · Personalized Quiz Generation

An end-to-end, deployed system covering objectives, methodology, evaluation results, and production deployment — built and evaluated across Milestones 1 through 6.

Group 3 - DS & AI Lab Project

Milestone 6

EXECUTIVE SUMMARY

One system, three intelligent capabilities, one deployed demo

117	9,427	0.93	100%
source documents processed (0 dropped)	retrievable chunks indexed for search	Precision@5 on the baseline RAG pipeline	duplicate-question detection precision

What we built: an end-to-end assistant that reads a Machine Learning course's transcripts, FAQs, notes and past-year questions, then (1) answers student questions with cited, grounded evidence, (2) detects and clusters duplicate/near-duplicate doubts so instructors see recurring confusion, and (3) generates personalized quizzes targeted at each student's weak topics.

- ✓ Grounded, cited answers — every claim traces to a source passage
- ✓ Recurring-doubt detection across FAQ, PYQ and practice-question banks
- ✓ Adaptive quizzes generated from each learner's flagged gap topics
- ✓ Containerized deployment: FastAPI + Qdrant + React, evaluated end-to-end

MOTIVATION

Course materials are rich – but finding the right answer inside them is slow

The problem

Students and instructors repeatedly search through long, fragmented lecture transcripts, FAQ pages, instructor notes and past-year question papers to find one specific answer.

The same doubts get asked again and again across FAQ threads, practice questions and exam papers — with no systematic way to see which topics genuinely trip students up.

Revision is generic: students rarely get quiz practice targeted specifically at the topics they are weakest on.

What the system aims to do

- Reduce search time with concise, direct answers instead of manual scanning
- Show the exact supporting passages behind every answer, for verification
- Surface recurring doubts across course materials so instructors can act on them
- Personalize revision by generating quizzes aimed at each student's gaps

WHO IT'S FOR

Primary use cases across three user groups

S

Students

- Quick revision without re-reading full transcripts
- Targeted Q&A during assignments and exam prep
- Citations they can use to verify or extend an answer

I

Instructors

- Visibility into the doubts students actually ask
- Identify ambiguous or under-explained topics
- Less time spent answering the same question repeatedly

M

Course Maintainers

- Automated indexing as new material is added
- A searchable, always-current knowledge base
- Structured logs for auditing answer quality over time

SOLUTION OVERVIEW

Three intelligent capabilities on one shared data foundation

Q&A

Retrieval-Augmented Q&A

- Dense + lexical retrieval over course chunks
- Cross-encoder reranking for top passages
- Grounded generation with explicit source citations
- Out-of-scope guardrail for irrelevant questions

QI

Question Intelligence

- Embeds every FAQ / PYQ / practice question
- Detects near-duplicate doubts (cosine + token guard)
- Clusters related doubts to surface recurring topics
- LLM-judged evaluation against Milestone 1 targets

Quiz

Personalized Quiz Engine

- Targets each learner's flagged weak topics
- Generates grounded multiple-choice questions
- Structural validation: 4 options, cited source chunks
- LLM-judged relevance scoring per question

Shared foundation: a single cleaned, chunked corpus (117 documents → 9,427 chunks) and embedding pipeline underpins all three capabilities.

DATA FOUNDATION

Corpus & scope

117
documents processed
(0 dropped, 0 errors)

9,427
chunks generated
for retrieval

420
question-bank units
(FAQ + PYQ + practice)

1.2%
duplicate rate found
in the question bank

Source material types

Lecture transcripts	Timestamped spoken-lecture text; cleaned of speaker tags and timestamps (4,949 removed)
FAQ documents	Curated staff-authored explainers — 143 question-bank units
Instructor notes / slides	Exported lecture notes and slide text used for grounding answers
Previous-year questions (PYQ)	OCR-extracted past exam papers — 155 question-bank units
Practice questions (PQ)	Course-authored practice sets — 122 question-bank units, 0% duplicate rate

DATA PREPARATION

Cleaning, chunking and leakage-free splitting

1

Clean

Unicode normalization (NFKC), whitespace/header cleanup, timestamp & speaker-tag removal

2

Chunk

Sliding-window chunking: ~400 tokens per chunk, 50-token overlap to avoid cutting sentences

3

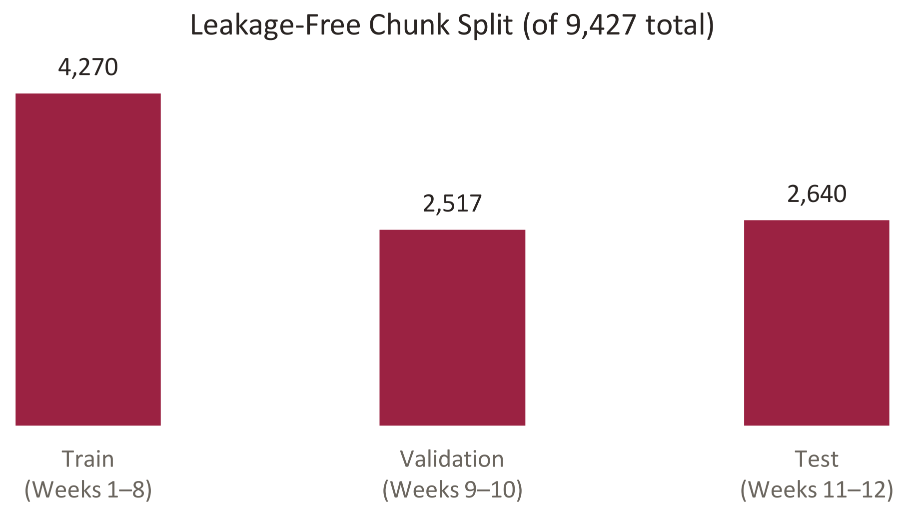
Split

Train / validation / test split by course week — leakage-free by construction

4

Index

Chunks embedded and pushed to the vector store, batched for scale



Cleaning pass — what actually needed fixing

Documents processed	117
Empty documents dropped	0
Files skipped (errors)	0
Timestamps removed	4,949
Duplicate headers removed	0
Unicode corruptions fixed	0

Embeddings, indexing and first-stage retrieval

1

Embed

Chunks converted to dense vectors: all-MiniLM-L6-v2 for RAG retrieval; BAAI/bge-small-en-v1.5 for question-intelligence dedup & clustering

2

Index

Qdrant (production, durable, payload metadata) or FAISS (fast local experiments: Flat / HNSW / IVF)

3

Retrieve

Top-k dense ANN search, optionally merged with BM25 lexical scores for exact-match / code / formula queries

Design choices & rationale

Embedding model	all-MiniLM-L6-v2 (RAG) — fast, compact, strong out-of-the-box baseline; larger models (all-mpnet-base-v2) available for ablation
Question-bank embeddings	BAAI/bge-small-en-v1.5 — used specifically for duplicate detection & clustering across FAQ / PYQ / PQ
Vector store	Qdrant for production durability & scaling; FAISS for fast local iteration during development
Hybrid scoring	$S = \alpha \cdot \mathrm{norm}(\mathrm{dense}) + (1-\alpha) \cdot \mathrm{norm}(\mathrm{BM25})$ — tunable weight to recover exact lexical matches dense search misses

ANSWER GENERATION

Reranking, grounded generation, and the out-of-scope guardrail

Pipeline

1. Retrieve top-k candidate passages (hybrid dense + BM25)
2. Cross-encoder reranker re-scores the top-10 for relevance (ms-marco-MiniLM-L-6-v2)
3. Prompt builder assembles ranked passages + question, instructing the model to use ONLY the supplied passages
4. Generator answers in a fixed template: direct answer, explanation, sources used
5. Guardrail check: if the question is out of the course's scope, the retrieval/generation path is skipped and a fixed refusal is returned

Grounding, in the prompt itself

"Use ONLY the passages below to answer."

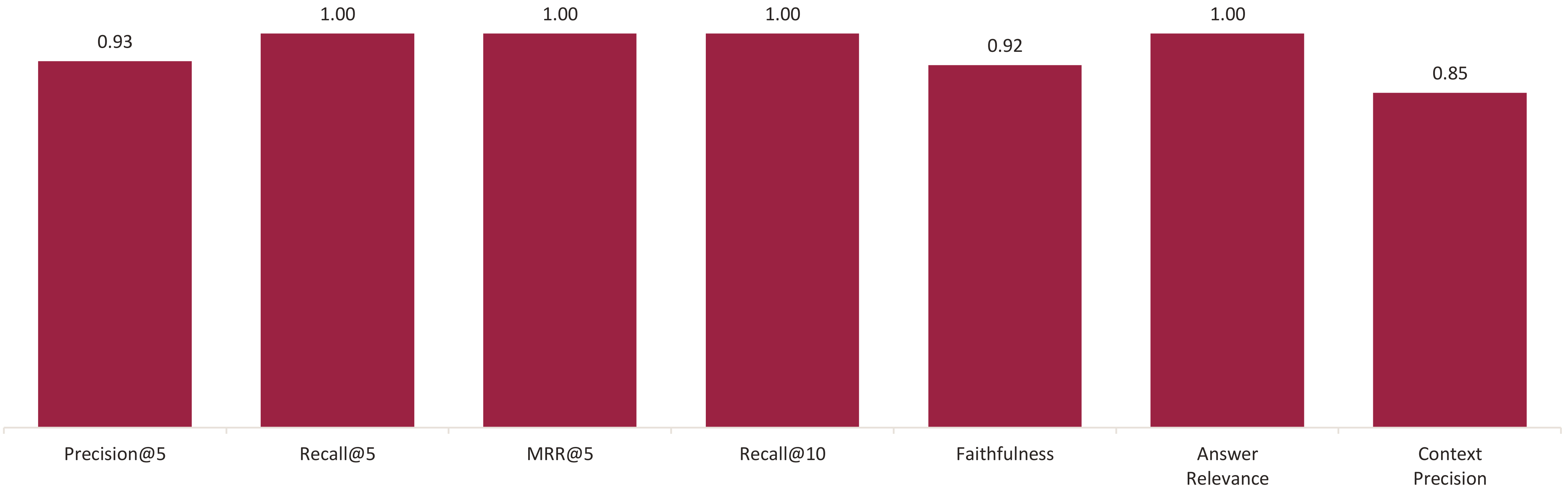
Every answer must list the passage IDs it used as sources, so a claim without a cited passage is visible as unsupported.

Out-of-scope guardrail — tested

Query: "What is quantum machine learning?" → correctly flagged out of scope; judged Faithfulness 1.00, Answer Relevance 1.00.

RAG EVALUATION

Retrieval & generation quality — aggregate scorecard



Evaluated on a sampled query set spanning in-scope conceptual, formula and worked-example questions plus an out-of-scope guardrail check.
Scores computed per query and averaged (see Slide 11 for two worked examples).

Two worked examples — an answered question and a guardrail refusal

IN SCOPE

"Explain the Bias-Variance tradeoff."

"Error = Bias + Variance. High bias (too simple) underfits; high variance (too complex) overfits. The goal is the balance that minimizes total error."

Sources used: [Context 1], [Context 2], [Context 3], [Context 4]

Faithfulness	**0.90**
Answer Relevance	**1.00**
Context Precision	**0.80**
Precision@5 / Recall@5	**1.00 / 1.00**

OUT OF SCOPE

"What is quantum machine learning?"

"This question is outside the scope of the ML course assistant. I can only answer questions about Machine Learning, AI, and related course topics."

No retrieval performed — the guardrail short-circuits before the RAG path runs.

Faithfulness	**1.00**
Answer Relevance	**1.00**
Context Precision	**1.00**

Detecting and clustering duplicate doubts

How it works

1. Every FAQ, PYQ and practice-question unit is embedded with BAAI/bge-small-en-v1.5
2. Duplicate flag: cosine similarity ≥ 0.95 AND a discriminative-token guard (numerals and polarity words must also agree)
3. Clustering: average-linkage on cosine distance, cut at 0.2, to group related-but-not-identical doubts
4. Evaluation judge: an independent model (groq/llama-3.3-70b-versatile) rather than the generator itself

Question bank under test

Units parsed	420
Distinct canonical doubts	415
Clusters formed	236
Duplicate rate	1.2%

By source type

Units → Canonicals → Dup. rate

PYQ	155 → 152	1.9%
FAQ	143 → 141	1.4%
PQ	122 → 122	0.0%

Generating quizzes targeted at each learner's gaps

1

Identify gap

Read the student's flagged weak topics from topic_recommendation_events

2

Retrieve context

Pull the relevant source_chunks for that topic from the indexed corpus

3

Generate MCQ

LLM (llama-3.3-70b-versatile via Groq) writes a 4-option question grounded in those chunks

4

Validate & judge

Structural checks (options, sourcing) plus an LLM-judge relevance score before serving

Sample generated question — "Kernel SVM", Week 10, medium

"In Kernel SVM, what replaces the data matrix X?"

A. ϕ(X) — marked correct

B. Y

C. X^TX

D. K

Sources: week10_notes_chunk_29, week10_faq_chunk_31

Judge scores (mean of all 3 sampled)

Tests the topic	1.00
Answerable from context	1.00
Exactly one correct option	1.00
Distractors plausible	0.80

All 3 sampled questions scored identically on these criteria.

QUIZ EVALUATION

Relevance, structural quality, and a regression check

100%

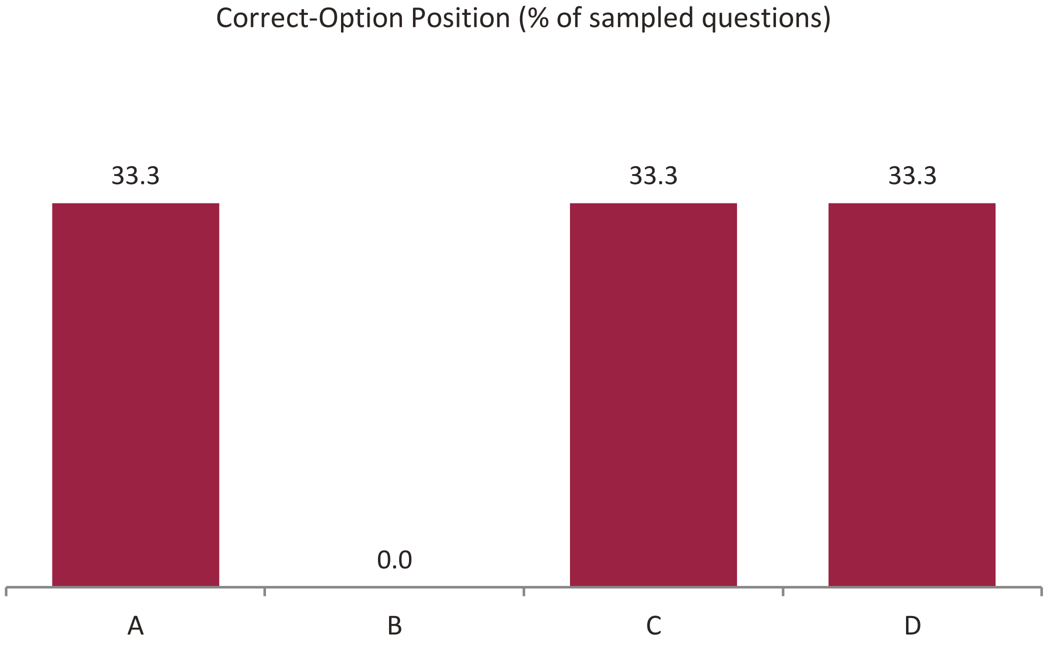
relevance to flagged weak topic
(target ≥ 80% — read as upper bound, see below)

100%

questions with non-empty
source_chunks and 4 distinct options

0 of 3

sampled questions citing the course's
own past-exam material (pq / PYQ)



Reading these numbers honestly

Regression check passed: answers now land roughly evenly across A/C/D (expected ~25% each), versus the old placeholder generator whose answer was always 'A'.

Relevance judge caveat: no independent judge model was reachable for this run, so the same model that generated the questions also graded them — the 100% figure is an upper bound, not an independent measurement. A second provider is needed before quoting it as final.

Pre/post revision improvement (target ≥ 15%): not yet reportable — no student/topic pair has ≥ 3 graded attempts on both sides of its first recommendation. Needs a seeded usage session.

ERROR ANALYSIS

What we measured, and what still needs work

Cluster recall is the weak link

Cluster pairwise F1 (6%) sits well below the 80% target — driven by low recall (3.1%) on a deliberately hard-negative-weighted sample; precision on flagged pairs is 70%.

Quiz relevance score is an upper bound

No independent judge model was reachable, so the generator graded its own quiz questions. Needs a second provider before the 100% figure can be trusted at face value.

Discourse (forum) data is absent, not stale

The forum source directory is empty in the current corpus — coverage is narrower than originally scoped. An admin can paste threads in manually; no scraper exists yet.

PYQ text is OCR output

Question boundaries aren't fully recoverable from scanned past papers, which hurts PYQ-specific duplicate detection and explains 0% observed citation of past-exam material in generated quizzes.

Revision-improvement metric needs real usage

No (student, topic) pair yet has ≥ 3 graded attempts on both sides of a recommendation — requires a seeded demo session to become measurable.

Known RAG failure modes

Truncated context at the token budget, domain-shift on lecture shorthand, and occasional unsupported claims — mitigated via chunk overlap, hybrid retrieval and strict grounding templates.

DEPLOYMENT

Architecture & technology stack

FRONTEND

Vite + React

Upload flow, preset examples, query box, cited-source answer view

API

FastAPI

/query · /upload · /health endpoints; stateless for horizontal scaling

VECTOR STORE

Qdrant / FAISS

Managed Qdrant in production; FAISS for local development

ORCHESTRATION

Docker Compose

API + Qdrant + frontend brought up with one command locally

Cloud & production notes

- Managed / hosted vector store in the cloud; QDRANT_URL points to the managed endpoint
- Secrets via a cloud secret manager — .env is never committed to the repository
- Stateless API instances behind a load balancer for horizontal autoscaling
- Reranker usage limited in production to bound latency; used more freely offline for evaluation
- Basic monitoring: request rate, average & p95 latency, error rate, and query volume logged
- Every generated answer writes an audit entry (timestamp, request id, model version, sources used)

LIVE DEMONSTRATION

We'll now present the deployed application directly

DEPLOYED APP URL

MLT Assistant

Demo flow we'll walk through:

1. Pick a preset example or upload a course document
2. Ask a question in plain language
3. Review the answer alongside its cited source passages
4. Try an out-of-scope question to see the guardrail respond
5. Open a personalized quiz generated from a sample weak topic

QUICK LOCAL START

```
docker-compose up -d
```

Brings up the API, Qdrant and the frontend together.

Health check

```
GET /health → 200 OK
```

LOOKING AHEAD

Future work & key takeaways

Future work

- Restore an independent judge model to re-score quiz relevance without generator bias
- Improve cluster recall on the hard-negative slice — the current gap to the 80% F1 target
- Build a discourse/forum ingestion path so recurring doubts there aren't missed
- Run a seeded usage session to make the pre/post revision-improvement metric reportable
- Domain adaptation for lecture-specific shorthand; richer comparative-results UI

Key takeaway

This project delivers an explainable, deployed learning assistant: grounded and cited Q&A, recurring-doubt intelligence across the question bank, and quizzes personalized to each learner's gaps — evaluated with real metrics, reported honestly including where targets were missed.

Thank you.